

Question

Metal **X** is soluble in aqua regia to obtain **A**, which is then treated with SO_2 to obtain **B**. Saturated solution of **Y** is added to the above mixture, and orange-red crystals of **C** precipitate. **C** can be reduced by H_2 under heating conditions to obtain elemental **X**. A dilute solution of **C** is colorless. When a concentrated solution of the highly toxic salt **Z** is added to it, a light yellow precipitate **D** precipitates. 1.00g of **D** can absorb 8.9mL (1atm, 0°C) of chlorine gas, resulting in a non-stoichiometric compound **E** with electrical conductivity, and the color changes from pale yellow to bronze. **E** can also be obtained by "combining" **D** and a certain complex **F** in solution in an approximate 5:1 ratio and precipitating. (The valence state of the metal in **F** is higher than that in **E**)

The crystallographic information of **C** and **Y** is as follows:

C is a tetragonal unit cell with three types of atoms. The atomic coordinates of all three types of atoms are as follows:

The first type of atom is located at (0, 0, 0)

The second type of atom is located at $(0, \frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, 0, \frac{1}{2})$

The third type of atom is located at $(\frac{1}{4}, \frac{1}{4}, 0)$, $(\frac{3}{4}, \frac{1}{4}, 0)$, $(\frac{1}{4}, \frac{3}{4}, 0)$, $(\frac{3}{4}, \frac{3}{4}, 0)$.

The unit cell parameters of **C** are $a = b = 702.4\text{pm}$, $c = 414.7\text{pm}$, and the density is 3.369 g/cm^3

Y has a rock salt structure with a unit cell parameter of $a = 629.2\text{pm}$ and a density of 1.987 g/cm^3

The crystal of **Z** is isostructural with **Y** at room temperature, and its hot water solution has a bitter almond taste.

Determine the chemical formulas of **X**, **Y**, **Z**, and **A**, **B**, **C**, **D**, **E**, and select the correct options from the following.

- A.** The sum of the reactant coefficients in the chemical equation for the dissolution of **X** in aqua regia (note: not the ionic equation) is 23.

B. The sum of the reactant coefficients in the chemical equation for the dissolution of **X** in aqua regia (note: not the ionic equation) is 24.

C. The sum of the reactant coefficients in the chemical equation for the dissolution of **X** in aqua regia (note: not the ionic equation) is 27.

D. In the crystal structure of **C**, the sum of the coordination numbers of the elements with positive valence is 10, and the difference in coordination numbers is 4.

E. In the crystal structure of **C**, the sum of the coordination numbers of the elements with positive valence is 16, and the difference in coordination numbers is 4.

F. In the chemical reaction equation where **E** is precipitated through **D** and a certain complex **F** in solution, the actual ratio of **D** to **F** is $\frac{17}{3}$.

G. In the chemical reaction equation where **E** is precipitated through **D** and a certain complex **F** in solution, the actual ratio of **D** to **F** is $\frac{16}{3}$.

H. None of the above options is correct

Answer

Correct Answer: **F**

Detailed Explanation

From $\rho = ZM/VN_A$

$$M(\mathbf{Y}) = \rho(\mathbf{Y})VN_A/Z = 1.987 \times 629.2^3 \times 10^{-30} \times 6.022 \times 10^{23}/4 = 74.5 \text{ g/mol}$$

Since $\mathbf{Y}$ is a 1:1 form of salt, it can be obtained through molar mass that: $\mathbf{Y}$ is KCl

CHECKPOINT

1 PTS

$\mathbf{Y}$ is KCl

$$M(\mathbf{C}) = \rho(\mathbf{C})VN_A/Z = 3.369 \times 702.4^2 \times 414.7 \times 10^{-30} \times 6.022 \times 10^{23}/1 = 415.1 \text{ g/mol}$$

According to the given proper unit cell and atomic coordinates of $\mathbf{C}$, the ratio of the three atoms can be obtained. Since $\mathbf{X}$ is a metallic element and usually does not have a negative valence, it can be inferred through valence states that $\mathbf{C}$ contains 1 X , 2 K , and 4 Cl .

Corresponding to the element mass, $\mathbf{X}$ is Pt , and $\mathbf{C}$ is K_2PtCl_4 .

CHECKPOINT

1 PTS

X is Pt

CHECKPOINT

1 PTS

C is K_2PtCl_4

Based on the description of bitter almond smell, it can be determined that **Z** is KCN , so **D** should be $K_2Pt(CN)_4$

CHECKPOINT

1 PTS

D is $K_2Pt(CN)_4$

According to the question, 1.00 mol **D** can absorb $377.4 \times 8.9 \times 10^{-3} / 22.4 = 0.15$ mol of chlorine gas, that is, 1.00 mol **D** contains 0.3 mol of Cl atoms. **E** is $K_2Pt(CN)_4Cl_{0.3}$

CHECKPOINT

1 PTS

E is $K_2Pt(CN)_4Cl_{0.3}$

Combining the information in other reactions, we get:

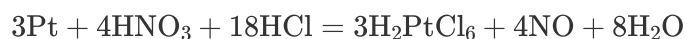
A is $K_2Pt(CN)_4$, **B** is $H_2Pt(CN)_4$, **C** is K_2PtCl_4

CHECKPOINT

1 PTS

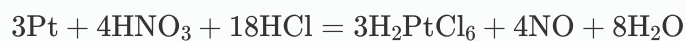
A is $K_2Pt(CN)_4$, **B** is $H_2Pt(CN)_4$

The chemical reaction equation for **X** dissolving in aqua regia is:



CHECKPOINT

1 PTS



The sum of the reactant coefficients is 25. Options A, B, and C are all incorrect.

CHECKPOINT

1 PTS

The sum of the reactant coefficients is 25. Options A, B, and C are all incorrect

From the given proper unit cell and atomic coordinates of **C**, it can be obtained that the coordination number of potassium is 8 and the coordination number of platinum is 4. That is, the sum of the coordination numbers is 12, and the difference in the coordination numbers is 4. Options D and E are both incorrect.

CHECKPOINT

1 PTS

The sum of the coordination numbers of the two elements is 12, and the difference in the coordination numbers is 4. Options D and E are both incorrect

E can also be "combined" from **D** and a certain complex **F** in solution in an approximate ratio of 5:1 and precipitated. The valence state of the metal in **F** is higher than that in **E**. The valence state of the metal in **E** is +2, and the valence state of the metal in **F** should be another common valence state of Pt, +4. According to "combination", it can be determined that **F** contains chloride ions, and then try to find a complex with a suitable ratio. It is obtained that **F** is $K_2Pt(CN)_4Cl_2$.

CHECKPOINT

1 PTS

F is $K_2Pt(CN)_4Cl_2$

The chemical reaction equation is:



CHECKPOINT

1 PTS



The actual ratio is $\frac{17}{3}$, option F is correct, and option G is incorrect.

CHECKPOINT

1 PTS

The actual ratio is $\frac{17}{3}$, option F is correct, and option G is incorrect.